Classical Field Theory — Exercise Sheet 22

Vortex orientation, scalar reduction and moduli

Hints and answers

Throughout this sheet $q = v = \lambda = 1$, $D_j = \partial_j - \mathrm{i}A_j$, $B = \partial_1 A_2 - \partial_2 A_1$, and the plane has orientation $\mathrm{d}x_1 \wedge \mathrm{d}x_2$. Use Lecture 22's energy and current $K_j = \frac{1}{2}\operatorname{Im}(\overline{\Phi} D_j \Phi)$. Every sector statement retains its finite-energy, uniform vacuum-modulus and absolute magnetic-integrability assumptions.

Exercise 22.1 (Reflection and higher-winding antivortices). Let $R(x_1, x_2) = (x_1, -x_2)$. For smooth fields set $\widetilde{\Phi} = \Phi \circ R$ and $\widetilde{A} = R^* A$.

(a) Track B, K, the energy and counterclockwise winding under this reflection. Derive the square completion for winding $-N$, $N \geq 1$, and prove the bound and equality equations in the reflected sector.

(b) For real radial profiles use $\Phi = f(r)e^{-\mathrm{i}N\theta}$ and $A = \alpha(r)\mathrm{d}\theta$. Derive both equality ODEs, the flux and the boundary conditions on the positive vacuum branch. Give sufficient smooth Cartesian core forms and the first corrections to $f = s_N r^N + \cdots$ and $\alpha = -r^2/4 + \cdots$. Here $s_N > 0$ is a local slope coefficient, not a proved global shooting value.

Hint. Reflection reverses curl and winding but not the energy integral. For the radial calculation use $\partial_1 - \mathrm{i}\partial_2 = e^{-\mathrm{i}\theta}(\partial_r - \mathrm{i}r^{-1}\partial_\theta)$. Integrate the logarithmic derivative of f/r^N near zero.

Answer. The negative-winding identity and equality equations are

$$e_1 = \frac{1}{2}|(D_1 - \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B + \frac{1 - |\Phi|^2}{2}\right]^2 - \frac{B}{2} - \operatorname{curl} K,$$

$$E_1 \geq \pi N, \qquad (D_1 - \mathrm{i}D_2)\Phi = 0, \quad B = -\frac{1}{2}(1 - |\Phi|^2).$$

The radial equations are $f' = (N + \alpha)f/r$ and $\alpha'/r = -(1 - f^2)/2$, with $\alpha(\infty) = -N$ and flux $-2\pi N$. The leading corrections are $f = s_N r^N(1 - r^2/8 + O(r^4))$ and $\alpha = -r^2/4 + s_N^2 r^{2N+2}/(4N + 4) + O(r^{2N+4})$.

Exercise 22.2 (The scalar equation with its point sources). Let a positive critical vortex in the stated sector have finitely many zeros $z_j \in \mathbb{R}^2$ of multiplicities $m_j \geq 1$. Away from the zeros put $w = \log|\Phi|^2$. Assume as local zero data that $w = 2m_j \log|x - z_j| + w_j$ near z_j, where w_j is smooth. This local regular remainder is a hypothesis, not a claim about arbitrary smooth fields with zeros. Use the distributional identity $\Delta \log|x - z_j| = 2\pi\delta_{z_j}$ as a given.

(a) On a phase patch derive A_1, A_2 in terms of w and the real phase. Eliminate the potential from the two Bogomolny equations.

(b) Prove the resulting equation on the whole plane as a distribution, including its point-source coefficient. Determine $\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x$ from the sector flux when $\sum_j m_j = N$. Does this reduction prove existence for prescribed zeros?

Hint. Write $\Phi = e^{w/2 + \mathrm{i}\theta}$ on a nonvanishing patch and separate the real and imaginary parts of $(D_1 + \mathrm{i}D_2)\Phi = 0$. The regular remainder lets the nonsingular part of the equation extend through each zero.

Answer. The equation and integral are

$$\Delta w + 1 - e^{w} = 4\pi \sum_{j} m_j \delta_{z_j}, \qquad \int_{\mathbb{R}^2} (1 - e^{w}) \mathrm{d}^2 x = 4\pi N.$$

They are consequences for the assumed vortex, not an existence proof.

Exercise 22.3 (Parameters at a two-vortex collision). A *moduli space* here is the set of critical positive N-vortex solutions with the vacuum asymptotics, modulo smooth gauge transformations. Use the following stronger geometric result as a *given*: this space is a nonsingular complex manifold, and the coefficients of

$$P(z) = \prod_{j=1}^{N} (z - z_j)$$

are smooth complex coordinates, including at coincident zeros. This input is not inferred from existence and uniqueness as a set alone. For its zero-coordinate description see Alonso Izquierdo, Garcia Fuertes, Manton and Mateos Guilarte, *JHEP* **01** (2024) 020, §5, pp. 10–11, equation (5.1) and discussion. Taubes's existence/uniqueness result is the one cited in Lecture 22.

(a) Count the real moduli and the moduli after fixing the mean zero position. Explain why ordering the zeros overcounts configurations.

(b) For $N = 2$, write $P(z) = z^2 - sz + p$. Replace (s, p) by the centre $Z = s/2$ and discriminant $\Delta = s^2 - 4p$. Locate the collision set and determine its real codimension. Explain why (z_1, z_2) and their separation are poor smooth coordinates there.

Hint. A monic degree-N polynomial has N free complex coefficients. For two zeros complete the square; do not choose a branch of a square root and assume that branch is smooth at zero.

Answer. There are $2N$ real moduli and $2N - 2$ after centring. For two vortices, $P(z) = (z - Z)^2 - \Delta/4$ and the collision set is $\Delta = 0$. It has real codimension two in the four-dimensional moduli space.

Exercise 22.4 (Designing a radial computation). For the positive unit-vortex ODEs $f' = (1-\alpha)f/r$ and $\alpha'/r = (1-f^2)/2$, design a computation on $[r_0, R]$, where $0 < r_0 \ll 1 < R$.

(a) Derive the coefficients through r^5 in f and through r^6 in α, using the free coefficient $s_0 = f'(0)$.

(b) Treat s_0 as an unknown parameter. Give three boundary residuals for a two-equation boundary-value solver using the core series and $f(R) = 1$. Explain why adding $\alpha(R) = 1$ independently would overdetermine this particular setup.

(c) Specify separate tests of origin truncation, outer-boundary truncation and solver accuracy. What further information is needed before a numerical residual could prove existence of a full-plane vortex? No numerical run is required in this problem.

Hint. Substitute odd and even power series in the two ODEs and match coefficients in increasing order. Count the free parameter along with the differential-equation order before choosing endpoint conditions.

Answer. The series are

$$f = s_0 r - \frac{s_0}{8} r^3 + \left(\frac{s_0}{128} + \frac{s_0^3}{32} \right) r^5 + O(r^7), \qquad \alpha = \frac{r^2}{4} - \frac{s_0^2 r^4}{8} + \frac{s_0^2 r^6}{48} + O(r^8).$$

Use both series at r_0 and $f(R) - 1$ as the three residuals. Refinement tests diagnose approximation errors but do not alone prove full-plane existence.